

Code Server — codeserver-1.0.0-dev-0.0.4

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Revision: 1 · **Last modified:** 2026-07-01T00:00:00Z

Fourth dev pre-release of **HelixCode**. This round hardens the **real-account SSH-key auth stack** shipped in codeserver-1.0.0-dev-0.0.3 into something an operator can actually reach and keep running, and it makes the **editor's extension (plugin) marketplace** a first-class, anti-bluff-tested capability. Three themes: (1) a **login reachability + usability fix** — an unauthenticated browser now lands on the login form instead of a bodyless "This page isn't working"; (2) **durability + host-safety hardening** — the whole stack (gate + editor + TLS edge) now survives session and full-host restarts, installs without the fragile npm path, and closes the --auth none loopback residual; (3) **marketplace coverage** — install / use / persist / configure real Open VSX extensions, proven LIVE, plus a VS Code Dark default theme.

Authoritative sources: user guide [docs/guides/AUTH.md](#), extensions guide [docs/guides/EXTENSIONS.md](#), edge-boot guide [docs/guides/EDGE_BOOT.md](#), feature ledger [docs/features/Status.md](#), prior release [codeserver-1.0.0-dev-0.0.3.md](#).

Fix — unauthenticated browsers reach the login form (75e2d9b)

- **Symptom.** Browsing the site root unauthenticated returned a **bodyless 401** (forward_auth denied with no login redirect) → Chrome rendered "**This page isn't working**" and the user never reached the login form.
- **Fix.** The gate's GET /auth now returns **303 → /login** for browser navigations (GET + Accept: text/html), which Caddy's forward_auth copies to the client so the browser lands on the login page. XHR / fetch / asset / API requests (Accept **without**

text/html) still receive a **bare 401**, so a programmatic caller is never served an HTML login page.

- **Root-caused via systematic-debugging (§11.4.102).** The log's transient 502s were separately proven to be **matrix-time chaos, not steady-state** (150 parallel authed requests → 150× 200, 0 failures).
- **Regression guards (§11.4.135).**
 - TestAuthBrowserNavigationRedirectsToLogin
 - TestAuthNonBrowserRequestsStay401 (Go, deterministic 3/3) + e2e_auth (B) hardened to assert **browser → 303 → /login AND api → 401** (live 5/5).

Hardening — durability, install robustness, host-safety (e0519bf)

Post-dev-0.0.3 hardening, subagent-driven, each landed with captured or sandbox-proven evidence:

- **Reboot-persistent TLS edge (deploy/quadlet/helixcode-caddy.container + scripts/install-edge-boot.sh + docs/guides/EDGE_BOOT.md).** A rootless-podman **Quadlet** so the Caddy edge survives a **full host reboot** — closing a real recurring outage where the compose edge's port-forwarder died with the launching session. The host-native gate + editor already persist via systemd --user linger; the Quadlet completes the trio. Dry-run-validated; the work caught + fixed a Quadlet EnvironmentFile boot bug. It does **not** auto-start (that would clash on 52443 with the live compose edge — operator opt-in).
- **Standalone-tarball code-server install (scripts/install-auth.sh).** The install switched to the **standalone GitHub-release tarball as PRIMARY** (npm was fragile — registry lag / ENOENT), with npm as fallback; §11.4.6 version-verified, idempotent, sandbox-proven (fetch + extract into a temp dir, the live install untouched).
- **UID-scoped loopback firewall (scripts/harden-loopback.sh + docs/scripts/harden-loopback.md + AUTH.md §5.1).** A UID-scoped loopback **OUTPUT** firewall (nft / iptables) closing the --auth none residual; --check / --apply / --remove, **refuses to guess the account, refuses to escalate**; shellcheck-clean; --check proven with no root.
- **§11.4.65 exports.** Synchronized .html + .pdf siblings for all release docs (real-text verified).

Feature — extension marketplace: install, use, persist, configure

Across 43e37a5, 849dc9a, 8351e4c, and the pending theme/popular-extensions landing, HelixCode's editor **extension (plugin)** capability becomes a first-class, LIVE-tested feature. The marketplace is **Open VSX** (code-server's default); the Microsoft VS Code Marketplace is licence-restricted to Microsoft products and is **not** used (stated as fact, no bypass — §11.4.6 / §11.4.112).

- **Anti-bluff install + use suite (43e37a5, tests/types/extensions_auth.sh, §11.4.169).** Proves the extension capability LIVE, all in throwaway mktemp dirs so the operator's live extensions-dir is **never mutated** (§11.4.14): **X1** marketplace reachable (Open VSX API 200 + real version, SKIP-with-reason if offline); **X2** install from marketplace (--install-extension redhat.vscode-yaml → exit 0, listed @version, valid on-disk package.json); **X3** usable + **LOADED** (a live throwaway code-server extension host actually loads it); **X4** cleanup (throwaway instance down, dirs trap-removed, LIVE extensions-dir fingerprint unchanged). **Live run PASS = 5/0.** Registered in run_all_types.sh risk-order (§11.4.132)
 - Challenge bank tests/banks/helixcode-extensions.yaml.
- **Operator extensions guide (849dc9a, docs/guides/EXTENSIONS.md, §11.4.99-sourced).** Honest guide to the extension system: default marketplace is Open VSX; MS Marketplace licence-restriction stated factually; install via UI + CLI (--install-extension / --list-extensions / --uninstall-extension) with the HelixCode --extensions-dir caveat; gallery config via product.json / EXTENSIONS_GALLERY (**the --extensions-gallery CLI flag does NOT exist in 4.117.0** — §11.4.6 verified); local .vsix; troubleshooting. Sources verified 2026-07-01. Plus scripts/show-extension-gallery.sh — a **read-only** probe printing the configured gallery (no mutation, never switches to MS Marketplace).
- **Login-form copy/paste clipboard buttons (8351e4c, §11.4.169/§11.4.170).** Accessible **copy** (sign-command + bare challenge) and **paste** (recognize armored SSH signature block(s); multi-match picker) icon buttons on the login form. Client module assets/login_enhance.js is a UMD recognizer (regex for BEGIN/END SSH SIGNATURE), clipboard capability-probed with graceful fallback, inserts as textarea .value **ONLY (XSS-safe)**, never auto-submits, progressive-enhancement (the page works JS-off). Proven by server_login_ui_test.go (Go PASS), tests/types/login_recognition.test.js (node unit 8/8), and tests/types/login_ui_auth.sh — L1 live served-page + L2 node unit + **L3 host-rendered Chromium/CDP pixel proof** (buttons render / labelled / non-overlapping + copy→clipboard + paste→fill + 2-sig→picker)
 - XSS-safe + no-auto-submit) — **LIVE PASS = 3/3** (§11.4.170 device-independent rendered-pixel proof).

- **UI-driven marketplace journey (8351e4c, tests/types/extensions_ui_auth.sh + bank).** Headless Chromium opens the Extensions view → searches Open VSX → clicks the real **Install** button, with pixel + OCR evidence — the real user-journey path (§11.4.48). **GREEN 5/1** with an honest **operator_attended SKIP** on install-completion (headless cross-origin Open VSX install hangs; the on-disk + host-load proof is the CLI sibling extensions_auth X2/X3 — §11.4.52, no faked PASS).
- **VS Code Dark default theme + popular-extensions coverage (pending commit — conductor to land; deploy/code-server/settings.default.json, tests/types/theme_default_auth.sh, tests/types/extensions_popular_auth.sh, tests/run_all_types.sh).** The seeded default settings set workbench.colorTheme to **Visual Studio Dark** (§11.4.162), which install_auth.sh copies to a fresh install's User/settings.json so every HelixCode deployment defaults to a VS Code dark theme; theme_default_auth.sh is the §11.4.135 regression guard (3/3). extensions_popular_auth.sh proves a curated set of **major popular** Open VSX extensions (Python / ESLint / Prettier / GitLens class) genuinely **install (P1) + use/load (P2) + persist across a fresh process (P3) + configure round-trip (P4)**, with the MS-proprietary set (Pylance / Live Share / Remote) confirmed **honestly ABSENT** from Open VSX (P5, licensing — §11.4.112), and the live extensions-dir left unmutated (P6).

Feature ledger reconciliation (eee6169, §11.4.153)

docs/features/Status.md + a new Status_Summary.md (§11.4.56) + HTML / PDF / DOCX exports (§11.4.153/§11.4.65), every row reconciled to a real file / evidence (§11.4.6/§11.4.123): auth surface **PASS** (23/23 matrix, Go 70 tests -race); login-redirect (75e2d9b) **PASS**; extension install+use **PASS** (extensions_auth 5/5); reboot-persistence **PENDING** (landed; full-reboot survival not yet captured); loopback firewall **OPERATOR-BLOCKED** (--apply needs root). **Video-confirmation is honestly PENDING on every row** — no real-use video corpus exists yet (NOT fabricated; PASS verdicts are backed by cited qa-results/** artefacts). Legacy container+password suites report an honest **SKIP**.

Durability

The whole stack is now systemd --user-managed and reboot-persistent:

Browser —HTTPS→ Caddy TLS edge — rootless Podman compose (live) + Quadlet (reboot-boot, opt-in)

```

└─ forward_auth → helix-auth    (host-native,
systemd --user, loopback:8081)
└─ reverse_proxy → code-server (host-native
AS milosvasic, systemd --user,
                                --auth none,
loopback:8080, seeded VS Code Dark default)

```

- **Gate + editor** persist across logout via `systemd --user linger`.
- **TLS edge** persists across a **full host reboot** via the rootless-podman **Quadlet** (`scripts/install-edge-boot.sh`, opt-in so it does not clash with the live compose edge on 52443).
- **Loopback --auth none residual** closable by the UID-scoped firewall (`scripts/harden-loopback.sh`, `root-gated --apply`).

Honest boundaries (§11.4.6)

- **Marketplace = Open VSX, not the MS VS Code Marketplace.** The Microsoft Marketplace is **licence-restricted to Microsoft products — Pylance / Live Share / Remote-* are unavailable** and confirmed ABSENT from Open VSX (documented, not a defect — §11.4.112). This is stated as fact; there is no bypass.
- **Public Let's Encrypt is operator-gated.** Publicly-trusted TLS needs a public domain + reachable :80/:443 or a DNS-01 token (LAN box); the ACME machinery is proven against a local CA (`docs/guides/TLS.md`).
- **--auth none on loopback:8080 residual is mitigated, not eliminated in the default deployment.** The gate/edge enforce all auth; a local host process reaching 127.0.0.1:8080 bypasses it. The **root-gated** `scripts/harden-loopback.sh --apply` closes it (a UID-scoped OUTPUT firewall); it stays **OPERATOR-BLOCKED** in the ledger until an operator runs it with root.
- **Reboot-persistent edge is opt-in.** The Quadlet does **not** auto-start (it would clash on 52443 with the live compose edge); full-host-reboot survival is **PENDING** in the ledger until captured on a real reboot.
- **UI-driven marketplace install-completion is operator_attended.** Headless cross-origin Open VSX install hangs; the autonomous proof of install+load is the CLI path (`extensions_auth X2/X3`) — no faked GUI PASS (§11.4.52).
- **No real-use video corpus yet.** §11.4.153 video-confirmation is honestly PENDING on every ledger row; PASS verdicts rest on cited `qa-results/**` captured evidence, never a fabricated video.
- **code-server pinned to 4.117.0** (newest on npm at release time; install now prefers the standalone tarball).

Testing (§11.4.169)

- **New / hardened suites** (tests/types/): `extensions_auth`, `extensions_ui_auth`, `login_ui_auth`, `login_recognition.test.js`, `theme_default_auth`, `extensions_popular_auth` — plus the hardened `e2e_auth` (B) browser-redirect regression assertion and the Go guards `TestAuthBrowserNavigationRedirectsToLogin` / `TestAuthNonBrowserRequestsStay401`.
- **Marketplace evidence path is anti-bluff (§11.4.69/§11.4.107)**: every PASS cites captured evidence — real install output, real on-disk manifests + entry files, real fresh-process listings, a real config round-trip, a real host-rendered pixel proof — never a metadata-only pass; all in throwaway dirs so the live `extensions-dir` is provably unmutated (§11.4.14 / §11.4.122).
- All new suites registered in `run_all_types.sh` risk-order (§11.4.132) with their Challenge banks.

LIVE VALIDATION AGGREGATE (release-gate run): full §11.4.169 test-type matrix GREEN — **29/29 suites PASS, 0 FAIL** (61 individual checks PASS · 0 FAIL · 24 honest SKIP-with-reason per §11.4.3 for absent topology/credentials/infra). Evidence: `qa-results/run_all/20260701T222753Z-2813294`. Feature suites: `login_ui_auth` 2/0/1 · `theme_default_auth` 3/0/0 · `theme_visual_auth` 2/0/1 (host-rendered pixel proof, §11.4.170) · `extensions_auth` 5/0/0 · `extensions_ui_auth` 5/0/1 · `extensions_popular_auth` 6/0/0. Three test-infra suites (`race`, `extensions_ui_auth`, `extensions_popular_auth`) were hardened against shared-host `ulimit -u` fork-pressure environmental false-FAILs (§11.4.1 / §11.4.3 / §11.4.174) — distinguishing genuine product defects (still FAIL) from host resource starvation (SKIP-with-reason) — and independently reviewed to a clean GO (§11.4.142 / §11.4.134, 5 findings F1-F4 + N1 all resolved); every §1.1 paired mutation re-proven to FAIL.

Migration

- **No breaking parameter changes from dev-0.0.3**. The SSH-key auth model, `HELIX_AUTH_*` parameters, and `PROJECTS_ROOT` are unchanged.
- **Reboot-persistent edge (optional)**: `scripts/install-edge-boot.sh` installs the rootless-podman Quadlet (opt-in; stop the live compose edge first to avoid a 52443 clash).
- **Loopback hardening (optional, root)**: `scripts/harden-loopback.sh --apply` closes the `--auth none` residual for the account.
- **Extensions**: install from Open VSX via the Extensions view or `code-server --install-extension <id>`; see `docs/guides/EXTENSIONS.md`.

- **Default theme:** fresh installs seed **Visual Studio Dark**; existing deployments keep their User/settings.json.